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Unit “ 4 : Session “ 9 : SLO “ 2 | Odd Man Out “ Pipelining Basics

■ I. Odd Man Out

1. Group Fetch, Decode, Execute, Write-back, Compiler ‘ Compiler “ software; the rest are pipeline stages.
2. Group Data hazard, Control hazard, Structural hazard, Cache miss ‘ Cache miss “ a memory-hierarchy event, not a pipeline hazard class.
3. Group Bypassing, Branch prediction, Operand forwarding, Interrupt handling ‘ Interrupt handling “ OS/exception mechanism; the others are hazard-mitigation techniques.
4. Group RISC pipeline, Superscalar, VLIW, Hard disk ‘ Hard disk “ storage device; the others are instruction-level-parallel architectures.
5. Group Pipeline register, ALU, Program Counter, Power supply ‘ Power supply “ not a processor datapath/control element.
6. Group Throughput, Latency, Speedup, Voltage ‘ Voltage “ an electrical property, not a performance metric.
7. Group 5-stage, 7-stage, 10-stage pipeline, Binary code ‘ Binary code “ a representation, not a pipeline configuration.
8. Group Bubble, Stall, Flush, Accelerate ‘ Accelerate “ the others are pipeline disruption terms.
9. Group Instruction-level parallelism, Data parallelism, Sequential execution ‘ Sequential execution “ the absence of parallelism.
10. Group CPI, IPC, MIPS, USB ‘ USB “ an I/O interface standard; the others are CPU performance metrics.